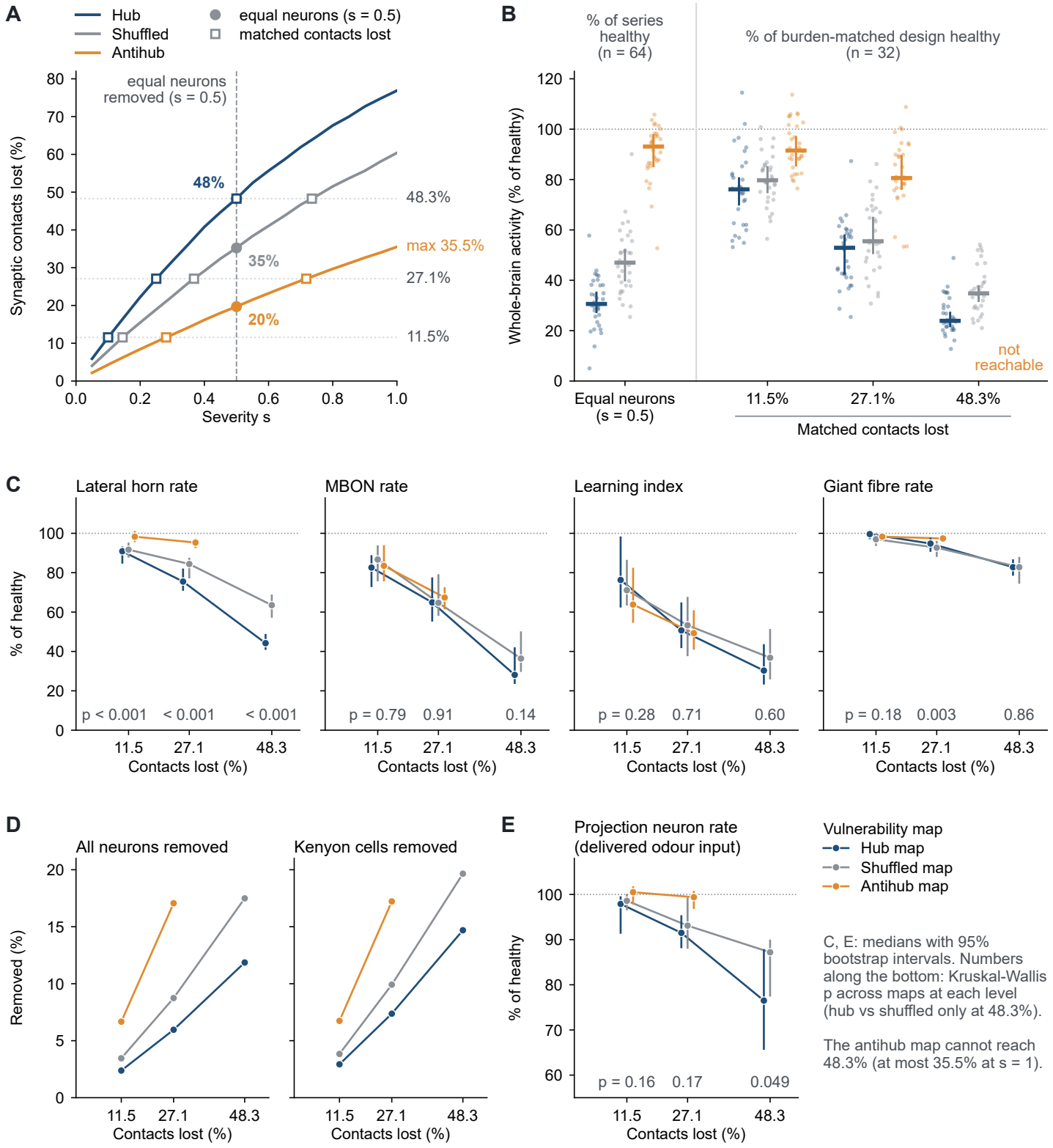

A
Hub
Shuffled
Antihub
equal neurons (s = 0.5)
matched contacts lost
equal neurons removed (s = 0.5)
48%
35%
20%
48.3%
max 35.5%
27.1%
11.5%
Synaptic contacts lost (%)
Severity s
B
% of series healthy (n = 64)
% of burden-matched design healthy (n = 32)
Whole-brain activity (% of healthy)
not reachable
Equal neurons (s = 0.5)
11.5%
27.1%
48.3%
Matched contacts lost
C
Lateral horn rate
MBON rate
Learning index
Giant fibre rate
% of healthy
p < 0.001 < 0.001 < 0.001
p = 0.79 0.91 0.14
p = 0.28 0.71 0.60
p = 0.18 0.003 0.86
Contacts lost (%)
D
All neurons removed
Kenyon cells removed
Removed (%)
Contacts lost (%)
E
Projection neuron rate (delivered odour input)
% of healthy
p = 0.16 0.17 0.049
Vulnerability map
Hub map
Shuffled map
Antihub map
C, E: medians with 95% bootstrap intervals. Numbers along the bottom: Kruskal-Wallis p across maps at each level (hub vs shuffled only at 48.3%).
The antihub map cannot reach 48.3% (at most 35.5% at s = 1).